

ORDER FLOW SOFTWARE

Functional Specification v2.0

Updated: orderflows.com indicators · Complete web reference list · Volume Profile as standalone instrument

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1. Complete Reference Registry

All sources consulted during the preparation of this specification — including the four PDF books provided by the user, the orderflows.com website, and all web sources independently researched — are catalogued below with their precise URLs.

1.1 Primary Source Documents (User-Supplied PDFs)

#	Title	Author	Year	Coverage
1	Trading Order Flow: Understanding & Profiting From Market Generated Information As It Occurs	Michael Valtos	2015	Footprint charts, Delta, COT, Imbalances, Stacked imbalance, Absorption, Trapped traders, Aggressive/passive participants, Initiative activity, VWAP, Iceberg orders, HFT, Candlestick + order flow, Chart types
2	ORDER FLOW: Trading Setups — Golden Ticket Edition	Dale (trader-dale.com)	2024	Volume clusters, HVN, Multiple nodes, Trades filter, Stacked imbalances, Unfinished business, Confirmation setups (big limit orders, absorption, aggressive orders, CVD divergence), Take profit/stop loss with order flow, Volume profile shapes, Volume profile setups
3	VOLUME PROFILE: The Insider's Guide to Trading — Golden Ticket Edition	Dale (trader-dale.com)	2024	Price action strategies, Volume profile POC/VAH/VAL, D/P/b/thin profile shapes, Volume accumulation setup, Trend setup, Rejection setup, Money management, Position sizing, Backtesting phases, Trading psychology
4	VWAP Book (vwap.com)	Dale (trader-dale.com)	2024	VWAP formula, Anchored VWAP (8 anchor types), VWAP strategies (reaction, rotation, trend), VWAP + Order Flow + Volume Profile confluences, Take profit placement, Stop loss placement, Trade trailing

1.2 Primary Website — orderflows.com (Newly Added)

 Fetched and fully read during preparation of v2.0. All pages listed below were directly accessed.

Page / Section	URL	Key Content Extracted
orderflows.com Home	https://www.orderflows.com/	Full product list, Pulse indicator summary, Turns indicator summary, Imbalance S&R summary, Single Prints summary, site navigation
Orderflows Pulse	https://www.orderflows.com/pulse.html	Full Pulse indicator specification: 7-variable composite (Order Flow, Delta, POC, Imbalances, Volume, Price Action, Swing Analysis) + Multi-Time Frame analysis upgrade, momentum jumps, order flow patterns
Orderflows Turns	https://www.orderflows.com/turns.html	Full Turns indicator specification: POC, bid/ask volume, volume, price action, delta, swing analysis. Works on footprint AND regular bar charts. Identifies turning points.
Imbalance Support & Resistance	https://www.orderflows.com/2015/08/22/imbalance-support-and-resistance/	Stacked imbalance zone indicator — draws buy/sell zones on chart, real-world analogy, 1–3 tick stops with 10x+ R/R potential
Top Heavy Big Seller / Bottom Heavy Big Buyer	https://www.orderflows.com/2015/08/22/big-buyer-seller/	Orderflows Ratio indicator — detects institutional big buyer at low or big seller at high. Acts as price floor/ceiling. Identifies institutional hedges and spreads.
Single Prints – Last Buyer / Last Seller	https://www.orderflows.com/2015/08/22/last-buyerseller/	Single print box indicator — marks last buyer at high / last seller at low. Signals end of move. Works on short and medium term charts.
Delta Surge (Free)	https://www.orderflows.com/deltasurge/index.html	Delta Surge indicator — analyzes delta to capitalize on real-time market information. Works on footprint AND normal bar charts. Free download. Delta trend analysis.
Orderflows Trader 7 Special	https://www.orderflows.com/special.html	Full list of 30+ automatable signals: Accumulation/Distribution, Open POC, Aligned POCs, Orderflows Gaps, Imbalance Reload, Volume Decline, POC Wave, Delta Tail, Resting Liquidity, Vertical Liquidity, Delta Breakout, Retail Suck, Price Action Divergence, Prominent POC, Orderflows Tails, Delta Divergence, Exhaustion Prints, Imbalance Reversals, Market Sweep, Market Weakness, Orderflows Sequencing, POC Slingshot, Zero Print, Engulfing Value Area, Inverse Imbalance, Stacked Imbalance, Thin Prints, Orderflows Ratio, Price Exhaustion,

		Price Defense, Multiple Imbalances
Orderflows Inner Circle Video Club	https://www.orderflows.com/videoclub/index.html	50+ webinar topics: Bar Delta Divergence (Webinar 18), 2nd Slot Imbalances (Webinar 19), Valtos U-Turn indicator (Webinar 23), Chart Types (Webinar 49), and many more.
Orderflows Toolbox 8.0	https://www.orderflows.com/toolbox/	Toolbox product — separate from NinjaTrader 8 basic Orderflows Toolkit. Lifetime license. No monthly fees.
Orderflows Software page	https://www.orderflows.com/software/	General software description, NinjaTrader integration details
Markets Traded	https://www.orderflows.com/markets-traded/	Works on any futures market with adequate volume: ES, NQ, YM, CL, GC, ZB, 6E, grains, energy

1.3 Web Sources Independently Researched (Not Supplied by User)

***i** All URLs below were found and read during research sessions — not provided by the user. These are additional reference sources used to build the specification.*

Source Name	URL	What It Contributed
NinjaTrader Official Blog — Order Flow / Footprint Charts (Oct 2025)	https://ninjatrade.com/futures/blogs/ninjatrade-order-flow/	Official definition of Footprint Chart vs Volumetric Bars; Order Flow+ suite description; Lee-Ready classification mention; industry-standard term confirmation
TradeThePool — Footprint Charts Explained (Oct 2025)	https://tradethepool.com/fundamental/footprint-charts-explained/	Visual definition of footprint profile; institutional interest zones; three-dimensional view of market activity
TradeThePool — Mastering Footprint Charts (Oct 2025)	https://tradethepool.com/fundamental/mastering-footprint-charts-trading/	Delta footprint definition; volume footprint definition; order flow footprint formal definition; unfinished auctions definition
LiteFinance — Order Flow Trading with Footprint Charts (Apr 2026)	https://www.litefinance.org/blog/for-beginners/trading-strategies/order-flow-trading-with-footprint-charts/	POC and Value Area definitions; passive vs aggressive order types; delta divergence as signal; delta trend analysis patterns
OptFutures — Footprint Charts Complete Guide	https://optimusfutures.com/blog/footprint-charts/	Three types of footprint charts (bid/ask, volume, delta); DOM explanation; stacked imbalances strategy; unfinished auctions strategy; image examples
Wikipedia — Order Flow Trading	https://en.wikipedia.org/wiki/Order_flow_trading	Buy/sell imbalance formal definition; spoofing/spoof orders definition; encyclopaedic reference for industry-

		standard terms
Triad City Beat — Footprint Charts for Beginners (Jan 2026)	https://triad-city-beat.com/footprint-charts-for-beginners-a-practical-guide-to-order-flow-trading/	Platform list (Sierra Chart, NinjaTrader, Bookmap, MT5); retail trader access to institutional tools; row-level price level description
TRADEPRO Academy — Footprint Chart Advantage	https://tradeproacademy.com/visualizing-market-order-flow-the-footprint-chart-advantage/	Range bar footprint example; WHO is in control framework; ES E-mini footprint chart example
GoCharting — Orderflows Trader User Guide (Apr 2026)	https://gocharting.com/docs/orderflow/orderflowtrader	Full Orderflows Trader platform documentation; stacked buying imbalance examples; delta candle divergence display; complete platform feature guide
ClusterDelta — Footprint for cTrader	https://clusterdelta.com/ctrader-footprint	Value Area Line Color settings; VAL/VAH line description; Profile BG inside/outside VA settings; cross-platform implementation reference
ClusterDelta — Footprint for MetaTrader 4/5	https://clusterdelta.com/footprint	Accumulation and distribution of market volumes; multi-platform compatibility reference; #Footprint indicator documentation
NordfX — Footprint Charts & Cumulative Delta (Nov 2025)	https://nordfx.com/en/useful-articles/footprint-charts-cumulative-delta-order-flow-trading	CVD divergence framework; combining footprint with classic price structure; demo account learning curve guidance; futures vs spot market correlation
Finowings — Order Flow Analysis: Footprint, Delta, Imbalance	https://www.finowings.com/Trading/order-flow-analysis-footprint-delta	Stacked imbalances support/resistance explanation; absorption definition; exhaustion definition; step-by-step order flow process for beginners
Wikipedia — NinjaTrader	https://en.wikipedia.org/wiki/NinjaTrader	TASC Readers Choice Award history (2021–2024); Kraken acquisition May 2025 for \$1.5B; platform history and brokerage info
Silver Creek Leadership — Mike Valtos Ultimate Bundle 30 (Feb 2026)	https://silvercreekleadership.academy/courses/orderflows-mike-valtos-ultimate-bundle-30/	Course overview; 30 modules; 2025 volatility spike updates; community Telegram group; ongoing update schedule
udcourse.com — Mike Valtos Bundle	https://udcourse.com/product/mike-valtos-bundle-30-orderflows-courses/	60–90 day screen time for tape reading mastery; Sierra Chart and NinjaTrader platform requirements; best markets (ES, NQ, CL, GC, ZB, BTC, ETH)
Orderflows Inner Circle Group	https://www.orderflows.com/innercircle/oldprice.html	Webinar 18 Bar Delta Divergence full definition; Webinar 19 2nd Slot Imbalances definition
orderflowstrader.com — Orderflows Trader 6	http://www.orderflowstrader.com/	8 new tools in v6: Thin Prints, Multiple Thin Prints, Delta/Volume Percentage, Extreme Delta/Volume, Extreme Delta,

		Small Min/Max Delta
Orderflows Trader 7 Special page	https://www.orderflows.com/special.html	Complete automatable signal list for v7; Markers Plus compatibility; third-party automation framework
Orderflows Toolbox page	https://www.orderflows.com/toolbox/	Toolbox vs Toolkit distinction; lifetime license model; no monthly fees policy
govinfo.gov — Citadel CEO Congressional Testimony (VWAP)	https://www.govinfo.gov/content/pkg/CHRG-117hhrg43966/pdf/CHRG-117hhrg43966.pdf	Official US Congressional testimony from Kenneth Griffin (Citadel CEO) confirming institutional VWAP program trading as virtually universal; VWAP slicing methodology
flowshifts.com — Michael Valtos background	http://www.flowshifts.com/	Confirmed 20-year institutional career: 8yr JP Morgan, 4yr Cargill, 3yr Commerzbank, 2yr EDF Man, 4yr Dean Witter Reynolds; aggressive vs passive framework explanation
TradingIQ — OrderFlow IQ (TradingView)	https://fr.tradingview.com/script/GX4WhZ8h-TradingIQ-OrderFlow-IQ	TradingView footprint implementation: POC, value area, imbalance detection, stacked imbalances, CVD levels across footprint

1.4 Reference Summary Count

Category	Count	Details
User-supplied PDF books	4	Valtos Trading Order Flow; Dale Order Flow Setups; Dale Volume Profile; Dale VWAP
orderflows.com pages (newly added)	12	Home, Pulse, Turns, Imbalance S&R, Big Buyer/Seller, Single Prints, Delta Surge, Trader 7, Inner Circle, Toolbox, Software, Markets
YouTube channel	1	@orderflows — channel metadata + video topic research
Independent web sources	23	NinjaTrader, TradeThePool ×2, LiteFinance, OptFutures, Wikipedia ×2, Triad City Beat, TRADEPRO, GoCharting, ClusterDelta ×2, NordfX, Finowings, Silver Creek, udcourse, orderflowstrader, govinfo.gov, flowshifts, TradingIQ, Inner Circle
TOTAL	40	All sources catalogued above

2. orderflows.com Indicators — Full Specification

The following indicators are Michael Valtos's proprietary products sold at orderflows.com. They are directly relevant to the software we are building — they define the signals and features that the commercial product must detect and surface. All specifications below are derived from the orderflows.com website pages fetched directly.

2.1 Orderflows Pulse

Source: <https://www.orderflows.com/pulse.html> — fetched directly

Description: The Orderflows Pulse is Valtos's most comprehensive composite indicator. It analyzes 7 order flow variables simultaneously and generates trading signals on both footprint charts AND regular candlestick charts. It received a major upgrade with Multi-Time Frame analysis, allowing higher-timeframe order flow signals to be overlaid on lower-timeframe charts.

2.1.1 The 7 Analysis Variables

Variable	What It Analyzes	Signal Contribution
Order Flow	What is currently trading and HOW it is trading in real-time	Identifies order flow patterns and execution aggression
Delta	Net difference between aggressive buyers and aggressive sellers	Shows which side is dominating the current bar
Point of Control (POC)	Price level with highest volume in the bar — only IMPORTANT POCs are used	Identifies institutional consensus price; POC slingshot and defense
Imbalances	Price levels where aggressive traders outnumber others by large percentages	Detects supply/demand imbalance; generates buy/sell signals
Volume	Sufficient volume threshold required for accurate assessment	Filters noise bars; only signals on bars with meaningful volume
Price Action	How price is reacting to the order flow context	Confirms or negates the order flow signal with price behavior
Swing Analysis	Reference to recent significant swing highs and swing lows	Provides context for current order flow relative to key market structure

2.1.2 Multi-Time Frame Analysis (Major Upgrade)

- Allows higher-timeframe order flow signals to be displayed on the current chart without switching timeframes
- Example: trading a 30-second ES chart — add signals from the 3-minute ES chart on top
- Requires the same 7-variable analysis to pass on BOTH timeframes for highest-conviction signal
- Works on any chart type: minute-based, volume-based, tick-based, range-based

2.1.3 Pulse Signal Types

Signal Type	Definition	Trading Application
Momentum Jump	Market momentum surges at a specific price — 'market sweet spot'. Optimal entry before and during trends.	Enter in direction of momentum jump; high-probability trend entry
Order Flow Pattern	Real-time analysis of how orders are executing — identifies immediate market dynamics	Confirms short-term directional bias based on live order activity
Delta Imbalance	Net aggressive buying vs selling dominance detected at the bar level	Identifies which side controls the bar; filters setups by direction
Volume Concentration	POC identification — highest volume price level in the bar	Finds institutional activity center; used for entry/stop placement
Price Imbalance	Aggressive trader concentration significantly outnumbers opposing side at a price level	Identifies likely support/resistance from the order flow data
Price Action Reaction	How price responds to the detected order flow conditions	Confirms or invalidates signal; filters false signals
Swing Reference	Context from recent swing highs and lows	Gives market structure context to order flow signal

2.1.4 Platform & Technical Details

- Runs on NinjaTrader 8 as an add-on indicator
- Install time: approximately 25 seconds
- Compatible with The Indicator Store's Markers Plus (third-party automation)
- Works on footprint charts (Orderflows Trader) AND regular bar/candlestick charts
- Works on minute, volume, tick, and range-based chart types with same settings
- Price: \$225 (one-time, 40% discount from \$397)

2.2 Orderflows Turns

Source: <https://www.orderflows.com/turns.html> — fetched directly

Description: The Orderflows Turns is designed specifically to identify market TURNING POINTS using a combination of 6 order flow and market analysis variables. It is simpler than the Pulse and more focused on reversal detection. Works on both footprint charts and normal bar/candlestick charts.

2.2.1 The 6 Analysis Variables

Variable	What It Analyzes
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Point of Control	Most-traded price within the bar — where institutional activity was concentrated
Volume (Bid & Ask)	Volume traded on bid side vs ask side — detects aggressive order dominance
Volume (Total)	Sufficient total volume threshold to validate signal quality
Price Action	How price is behaving relative to the order flow — confirms or denies the potential turn
Delta	Net buyer vs seller aggression — is the turn confirmed by delta reversal?
Swing Analysis	Recent swing high/low reference points — is price at a significant structural level?

2.2.2 Turns Signal Behavior

- Identifies areas of the chart being defended or supported by institutional activity
- Like having Valtos 'looking over your shoulder' pointing out defended/supported areas
- NOT a simple price pattern — combines multiple order flow data points
- Can be overlaid on top of existing footprint charts OR existing candlestick charts
- Designed for short-term day traders: recommended 4–10 range charts, timeframes under 5 minutes to 30 seconds
- Works on: ES, CL, YM, NQ, 6E, ZB, FGBL and all major liquid futures markets
- Not recommended for thinly-traded markets (Lean Hogs, Live Cattle)
- Platform: NinjaTrader 8 version 15+. Price: \$500 one-time.

2.3 Orderflows Ratio — Top Heavy Big Seller / Bottom Heavy Big Buyer

Source: <https://www.orderflows.com/2015/08/22/big-buyer-seller/> — fetched directly

Description: The Orderflows Ratio indicator detects the presence of institutional players so large that their activity creates short-term price floors and ceilings. Based on Valtos's personal experience placing 1000+ lot institutional orders.

2.3.1 Signal Logic

- Bottom Heavy Big Buyer: detected at a LOW — a large institutional buyer absorbs all selling at a price level, acting as a price FLOOR
- Top Heavy Big Seller: detected at a HIGH — a large institutional seller satisfies all buyers at a price level, acting as a price CEILING
- Based on bid/ask RATIO across the bar — the ask/bid ratio exceeds a configured threshold indicating one side is massively dominant
- May represent commercial hedges or spread trades — Valtos's specific use case from his institutional trading career

2.3.2 Trading Application

- When Bottom Heavy Big Buyer appears at a low: buy the retest; stop below the detected level
- When Top Heavy Big Seller appears at a high: sell the retest; stop above the detected level
- 1–3 tick stops with profit potential 10x+ the risk (Valtos's stated specification)
- Works on the same markets as other Orderflows indicators — any liquid futures

2.4 Single Prints — Last Buyer / Last Seller

Source: <https://www.orderflows.com/2015/08/22/last-buyerseller/> — fetched directly

Description: The Single Print Box indicator identifies when the LAST buyer has bought at a high or the LAST seller has sold at a low. This marks the potential END of a trend leg — nobody wants to transact at that extreme anymore.

2.4.1 Signal Logic

- Last Buyer at High: only 1–2 trades executed at the high — demand has dried up at that price. Price will begin to fall.
- Last Seller at Low: only 1–2 trades at the low — supply has dried up at that price. Price will begin to rise.
- The single print (low/zero volume at extreme) is the visual indicator on the footprint chart
- Works as both scalp (short-term) and swing (medium-term) trade signal
- The indicator draws a box around the single print for easy visual identification

***i** This is the orderflows.com implementation of what Dale calls 'Unfinished Business' / 'Failed Auction'. Same concept, different naming convention.*

2.5 Delta Surge (Free Indicator)

Source: <https://www.orderflows.com/deltasurge/index.html> — fetched directly

Description: Delta Surge is a FREE indicator from Valtos that analyzes order flow delta and works on BOTH footprint charts AND normal bar charts. Its purpose is to help traders who are not yet using footprint charts to access delta analysis.

2.5.1 Signal Logic

- Analyzes delta trend — in a healthy uptrend, delta builds on itself as new buyers are drawn in
- Detects surges in delta that signal momentum acceleration in the prevailing direction
- Works independently of footprint chart — reads delta data from the instrument feed
- Allows delta analysis on standard candlestick charts without requiring Orderflows Trader
- Download: free at <https://www.orderflows.com/deltasurge/index.html>

- Use case: first step for traders transitioning from traditional charts to order flow analysis

2.6 Orderflows Indicators — Complete Feature Matrix for Implementation

The following table maps all orderflows.com indicators to required software features and signals the C++ engine must support:

Indicator	Required Engine Signals	Output to API	Priority
Pulse	Delta, POC, Imbalances, Volume threshold, Price action context, Swing H/L, Multi-TF analysis	REST: /v1/signals/pulse WS: signals stream	P1 — v1.0
Turns	POC, Bid/Ask ratio, Volume, Price action, Delta, Swing H/L, Turning point detection	REST: /v1/signals/turns WS: signals stream	P1 — v1.0
Orderflows Ratio (Big Buyer/Seller)	Bar-level bid/ask ratio exceeding threshold at high/low; volume confirmation	REST: /v1/signals/ratio WS: signals stream	P1 — v1.0
Single Prints (Last Buyer/Seller)	Zero or near-zero volume at bar extreme; single print box detection	REST: /v1/signals/single_prints WS: signals stream	P1 — v1.0
Delta Surge	Delta trend acceleration; delta surge threshold detection; works on all chart types	REST: /v1/signals/delta_surge WS: delta stream	P1 — v1.0
Stacked Imbalance Zones	3+ consecutive imbalances; zone drawing; persistence across bars	REST: /v1/zones WS: signals stream	P1 — v1.0
All Trader 7 signals (30+)	Full signal set per Trader 7 spec (Accumulation, POC Wave, Delta Tail, etc.)	REST: /v1/signals WS: signals stream	P2 — v1.5

3. Volume Profile as a Standalone Trading Instrument

Volume Profile is not merely a supporting indicator — it is a complete, standalone analytical instrument with its own data model, display specification, trading setups, and API surface. This section specifies Volume Profile as a first-class instrument in the software.

3.1 What Makes Volume Profile a Standalone Instrument

- Volume Profile provides the STRUCTURAL CONTEXT within which all order flow signals are interpreted
- Without Volume Profile, order flow signals have no reference frame — a stacked imbalance in the middle of a value area means something different from one at the POC
- Institutional traders rely on Volume Profile (Market Profile) to identify fair value, value area, and initiative vs responsive activity levels
- Dale explicitly states: 'I don't look for candlestick formations... I perceive the price movement as a continuous flow' — Volume Profile IS that continuous flow framework
- Volume Profile can be traded completely independently of order flow footprint charts — Dale's original approach before adding order flow

3.2 Volume Profile Data Model

Data Element	Type	Calculation	Significance
POC — Point of Control	Price level	Price with highest total volume in the profile period	Most-traded price = institutional consensus of fair value. Strongest S&R level.
VAH — Value Area High	Price level	Upper boundary of 70% volume range above POC	Upper edge of institutional comfort zone. Strong resistance.
VAL — Value Area Low	Price level	Lower boundary of 70% volume range below POC	Lower edge of institutional comfort zone. Strong support.
Value Area	Price range [VAL, VAH]	70% of all traded volume falls within this range	Zone of price acceptance. Two-sided trade. Most likely for price to return here.
HVN — High Volume Node	Price cluster	Volume significantly > average for adjacent levels	Institutional accumulation/distribution zone. Strong S&R. Price spends time here.
LVN — Low Volume Node	Price gap	Volume significantly < average for adjacent levels	Price acceptance gap. Price moves quickly through these levels.
Volume Distribution	Histogram	Volume at each price level plotted horizontally	Full picture of institutional activity across all prices in

			the period
Bid/Ask Split Profile	Dual histogram	Ask volume (buy) vs Bid volume (sell) per price level	Shows directional bias within the profile — which side is more aggressive at each price

3.3 Volume Profile Types (All Must Be Implemented)

Profile Type	Period Definition	Reset Behavior	Primary Use Case	Timeframe
Session Profile	Single RTH session (e.g., 9:30am–4:00pm ET)	Resets each session open	Intraday trading — identifies today's fair value and value area	1min–30min
Daily Profile	One calendar day (including extended hours)	Resets at midnight	Daily structure analysis; previous day POC as reference level	5min–1hr
Weekly Profile	Monday open to Friday close	Resets each Monday	Swing trading context; weekly fair value benchmark	30min–4hr
Monthly Profile	Calendar month	Resets each month	Institutional positioning; long-term supply/demand map	Daily
Yearly Profile	January 1 to December 31	Resets each January 1	Annual fair value; VWAP alignment analysis	Daily–Weekly
Flexible / Anchored Profile	User-defined start date to current bar	Manual anchor — does not reset	Custom analysis: earnings events, regime changes, macro news	Any
Composite Profile	Merged across multiple sessions	Does not reset — accumulates	Long-term institutional level mapping; multi-day absorption zones	Any
Real-Time Profile	Current session building in real time	Continuous update each tick	Live trading — watch POC migration, value area expansion	Any

3.4 Volume Profile Shape Classification Engine

The engine must classify the current profile shape automatically and expose the classification via API. Shape determines the expected market behavior.

Shape Name	Visual Pattern	Mathematical Definition	Market Interpretation	Expected Next Move
D-Profile (Bell	Wide symmetric	Standard deviation of	Balanced market. Two-	Continue

Curve)	distribution centered on POC	volume distribution is low; POC is near the median price; tail volume < 15%	sided trade. Neither buyers nor sellers dominating.	rotation within value area until imbalance breaks out
P-Profile (Buy Imbalance)	Narrow base, wide top; POC is near upper 30% of range	Volume skewed to upper prices; lower prices have thin volume tail	Strong buying occurred near highs; accumulation at lows early, distribution later.	Continuation UP if holding above POC; short on failure below POC
b-Profile (Sell Imbalance)	Wide base, narrow top; POC is near lower 30% of range	Volume skewed to lower prices; upper prices have thin volume tail	Strong selling occurred near lows; distribution at highs early, accumulation at lows.	Continuation DOWN if holding below POC; long on failure above POC
Thin Profile	Narrow, elongated; minimal volume at most levels	Standard deviation of volume distribution is high; no clear POC concentration	Trending/initiative market. Price moved quickly. Little consensus.	Fast reversion to value if move is responsive; fast continuation if initiative

3.5 Volume Profile Trading Setups (Standalone)

Source: Dale — Volume Profile Insider's Guide; Dale — Order Flow Trading Setups ch. Volume Profile

Setup VP-1: Volume Accumulation Setup (D-Profile Breakout)

- Identify a D-profile (balanced, bell-curve) area — price has been rotating inside value area
- Wait for price to break OUT of the value area with conviction (bar closing clearly above VAH or below VAL)
- Entry trigger: first bar closing outside value area with delta 3x or more above average
- Stop: back inside the value area (VAH becomes support for long; VAL becomes resistance for short)
- Target: next significant HVN, POC, or VWAP level beyond the breakout
- Logic: the D-profile represents institutional accumulation — breakout signals their distribution/initiative

Setup VP-2: Trend Setup (P/b-Profile Continuation)

- Identify a P-profile (bullish) or b-profile (bearish) — shows directional institutional intent
- P-profile: wait for pullback to POC or VAH from above; look for long entry
- b-profile: wait for rally to POC or VAL from below; look for short entry

- Entry trigger: delta confirmation at the pullback level; delta should be in trend direction
- Stop: clearly on wrong side of POC (below for longs in P-profile; above for shorts in b-profile)
- Target: previous swing high/low; next week's VWAP; next profile POC
- Logic: POC is institutional consensus — they will defend it in trend direction

Setup VP-3: Rejection Setup (Thin Profile Reversal)

- Identify a thin profile area — price moved quickly through with little volume = no institutional acceptance
- Price reaches the thin profile extreme and shows aggressive rejection delta
- Entry trigger: rejection bar with strong opposing delta; CVD divergence at thin profile extreme
- Stop: beyond the thin profile extreme (1–2 ticks past the zero-volume zone)
- Target: back to the origin of the thin profile (nearest HVN or previous session POC)
- Logic: institutions did not transact here — no support/resistance. Price snaps back to value.

Setup VP-4: Value Area Re-Entry (Responsive Trade)

- Price breaks BELOW VAL or ABOVE VAH but with weak or negative delta — responsive move
- Price rapidly reverses back inside the value area
- Entry: on return into value area at VAL/VAH with confirming order flow
- Stop: beyond the recent extreme (outside the value area)
- Target: POC (value area center) — responsive trades target a return to fair value
- Logic: institutions did not support the breakout — they faded it with limit orders

Setup VP-5: POC as Dynamic Support/Resistance

- The POC of the previous session/day/week acts as a reference level for the next session
- Previous Session POC (VPOC): if current price is above, acts as support; if below, acts as resistance
- Strong signal when current session volume builds around the previous VPOC — institutional memory
- Entry: test of previous VPOC with order flow confirmation (absorption or buying imbalance at support VPOC)
- Stop: break below previous VPOC with strong sell delta
- Target: current session VAH or VAL

Setup VP-6: Volume Profile + Order Flow Confluence (Highest Conviction)

- Find a Volume Profile key level (POC, VAH, VAL, HVN) — this is the structural setup

- Wait for price to arrive at the level
- Require order flow confirmation: stacked imbalance OR absorption OR delta divergence at the level
- Also require VWAP alignment: price at VP level that also coincides with Daily or Weekly VWAP
- Triple confluence (VP + Order Flow + VWAP) = highest conviction trade signal in the system
- Entry: at the confluence level; stop 1–3 ticks outside; target 2:1 or better R/R to next structure

***i** This is the core methodology that both Valtos and Dale describe as their preferred approach. Three-way confluence is the gold standard.*

3.6 Volume Profile API Surface

Endpoint	Method	Parameters	Response
/v1/profile/{instrument_id}	GET	period_type, start_date, end_date, session_filter	{ poc, vah, val, value_area_pct, hvns[], lvns[], shape, distribution[] }
/v1/profile/{instrument_id}/live	GET	period_type	Real-time current session profile building
/v1/profile/{instrument_id}/composite	GET	from_date, to_date	Composite profile across multiple sessions
/v1/profile/{instrument_id}/anchored	GET	anchor_timestamp	Flexible anchored profile from any point to current bar
WS: profile/{instrument}	WebSoc ket	Subscribe to profile type	Live updates: { poc, vah, val, shape, updated_at } on each bar close

3.7 Volume Profile Display Specification

Display Element	Default Color	Configurable	Notes
Volume histogram bars	Blue gradient (ask-heavy) / Red gradient (bid-heavy)	Yes	Darker = higher volume. Optional bid/ask split coloring.
POC line	Yellow horizontal line across chart	Yes	Most prominent line on the profile
VAH line	Green dashed horizontal line	Yes	Upper value area boundary
VAL line	Red dashed horizontal line	Yes	Lower value area boundary
Value area fill	Semi-transparent blue	Yes	Shaded region between VAL and VAH

HVN markers	Bold/wider volume bar	Yes	HVN automatically detected and highlighted
LVN markers	Thin/barely visible gap	Yes	LVN gaps automatically highlighted
Shape label	Text overlay: D / P / b / Thin	Yes	Auto-classified shape label displayed on chart
Profile type label	Text: Session / Daily / Weekly / etc.	Yes	Shows which profile type is displayed

3.8 Volume Profile Configuration Parameters

Parameter	Default	Range	Description
value_area_percent	70	60–90%	Percentage of volume defining value area (academic standard is 70%)
profile_type	SESSION	SESSION/ DAILY/ WEEKLY/ MONTHLY/ YEARLY/ FLEXIBLE/ COMPOSITE	Which profile type to display
show_multiple_profiles	false	true/false	Show multiple profiles simultaneously (e.g., session + weekly)
hvn_threshold_multiplier	1.5	1.1–3.0	Volume must exceed average × multiplier to qualify as HVN
lvn_threshold_multiplier	0.5	0.1–0.9	Volume must be less than average × multiplier to qualify as LVN
bid_ask_split	false	true/false	Show bid and ask volume separately in profile histogram
profile_width_pct	20	5–50%	Profile histogram width as percentage of chart width
auto_classify_shape	true	true/false	Auto-detect and label D/P/b/Thin profile shape
show_previous_poc	true	true/false	Display previous session/period POC as reference line
composite_lookback_days	20	5–252	Number of days to include in composite profile
real_time_update_ms	100	100–5000ms	How often live profile recalculates during session

4. Updated Complete Feature Matrix

Full list of all features across Order Flow, Volume Profile, VWAP, and orderflows.com Indicators that the software engine must implement.

Feature Category	Feature	Source	Version
Footprint	Bid/Ask volume per price level	Valtos book; Dale book; NinjaTrader	v1.0
Footprint	Cell color (green/red) by ask vs bid dominance	Valtos; Dale	v1.0
Footprint	Volume heat map (darker = higher)	Dale	v1.0
Footprint	Diagonal imbalance comparison	Valtos; Dale	v1.0
Delta	Bar delta (ask_vol - bid_vol)	Valtos; Dale; GoCharting	v1.0
Delta	Max delta / Min delta within bar	Valtos	v1.0
Delta	Cumulative Delta (CVD) — session total	Dale; GoCharting	v1.0
Delta	Delta/Volume percentage	Orderflows Trader 6	v1.0
Delta	Extreme delta detection	Orderflows Trader 6	v1.0
Delta	Small Min/Max delta detection	Orderflows Trader 6	v1.0
Delta	Delta Surge indicator	orderflows.com/deltasurge	v1.0
Delta Divergence	Standard: negative delta at new high / positive at new low	Valtos; Dale	v1.0
Delta Divergence	Bar Delta Divergence: bearish bar + positive delta	Valtos Webinar 18	v1.0
Delta Divergence	CVD Divergence: price vs cumulative delta	Dale	v1.0
Imbalance	Single buying/selling imbalance (2.5x/3x/4x)	Valtos; Dale; OptFutures	v1.0
Imbalance	Stacked imbalance (3+) zone with zone drawing	Valtos; Dale; orderflows.com	v1.0
Imbalance	2nd Slot imbalance detection	Valtos Webinar 19	v1.0
Imbalance	Imbalance reversal	Orderflows Trader 7	v1.0
Imbalance	Multiple imbalances cluster	Orderflows Trader 6/7	v1.0
Imbalance	Inverse imbalance	Orderflows Trader 6	v1.0
Imbalance	Imbalance reload	Orderflows Trader 7	v1.0
COT	Intrabar COT: highest volume price level in bar with box display	Valtos book	v1.0

Absorption	High volume + low net delta at price level	Valtos; Dale	v1.0
Trapped Traders	Imbalance at bar extreme in opposite direction to bar close	Valtos book	v1.0
Initiative/ Responsive	Price vs value area + delta at breakout	Valtos book	v1.0
Single Prints	Last buyer at high / last seller at low detection	orderflows.com/last-buyerseller	v1.0
Orderflows Ratio	Top heavy big seller / bottom heavy big buyer	orderflows.com/big-buyer-seller	v1.0
Orderflows Pulse	7-variable composite signal + multi-TF analysis	orderflows.com/pulse.html	v1.0
Orderflows Turns	6-variable turning point detector	orderflows.com/turns.html	v1.0
Volume Profile	Session profile with POC/VAH/VAL/HVN/LVN	Dale VP book; orderflows.com	v1.0
Volume Profile	Daily/Weekly/Monthly/Yearly profiles	Dale VP book	v1.0
Volume Profile	Flexible / Anchored profile	Dale VP book	v1.0
Volume Profile	Composite profile (multi-session)	Dale VP book	v1.0
Volume Profile	D/P/b/Thin shape auto-classification	Dale VP book	v1.0
Volume Profile	Prominent POC + Aligned POCs + Open POC	Orderflows Trader 7	v1.0
Volume Profile	POC Wave + POC Slingshot	Orderflows Trader 7	v1.0
Volume Profile	Bid/Ask split profile histogram	Dale book	v1.0
Volume Profile	VP Setup 1–6 signal detection	Dale; this spec §3.5	v1.0
VWAP	Daily/Weekly/Yearly VWAP	Dale VWAP book	v1.0
VWAP	Anchored VWAP (8 anchor types)	Dale VWAP book	v1.0
VWAP	Standard deviation bands (1σ , 2σ)	Dale VWAP book	v1.0
VWAP	VWAP reaction/rotation/trend setups	Dale VWAP book	v1.0
Trader 7 Signals	All 30+ automatable signals (Accumulation thru Multiple Imbalances)	Orderflows Trader 7 spec	v1.5
Market Structure	Session high/low, previous day high/low	Valtos book	v1.0
Market Structure	Unfinished Business / Failed Auction detector	Dale book; orderflows.com	v1.0
Chart Types	TIME / VOLUME / RANGE bar construction	Valtos book	v1.0

5. Implementation Notes — v2.0 Additions

5.1 Volume Profile as Primary Data Layer

Volume Profile data must be calculated and available BEFORE order flow signals are evaluated. The signal engine depends on VP context for signal validity:

- Stacked imbalance at VAH is stronger than stacked imbalance in middle of value area
- Delta divergence AT the POC is a high-conviction signal; delta divergence away from POC is lower conviction
- Absorption at the POC has different meaning than absorption at an HVN vs an LVN
- All 6 VP trading setups must be cross-referenced with order flow signals before alerting the user

5.2 orderflows.com Indicator Compatibility

 *The Pulse and Turns indicators are sold as commercial NinjaTrader products by Valtos. We are building our OWN implementation of these signal types, not reverse-engineering his protected code. Our signals follow the same analytical logic as documented on orderflows.com.*

- Pulse implementation: combine all 7 variables in a weighted scoring engine; threshold score triggers signal
- Turns implementation: 6-variable analysis with turning point pattern recognition
- Ratio implementation: bar-level bid/ask ratio at price extremes with configurable threshold
- Single Prints implementation: zero-print / last-print detection at bar highs/lows
- Delta Surge implementation: delta acceleration detection using rolling delta window

5.3 Multi-Instrument Volume Profile

Per Valtos's teaching: the same Volume Profile concepts apply across ALL futures markets. The software must support configuring profiles for:

- Index futures: ES (S&P 500), NQ (Nasdaq 100), YM (Dow Jones), RTY (Russell 2000)
- Energy: CL (Crude Oil), NG (Natural Gas), RB (RBOB Gasoline)
- Metals: GC (Gold), SI (Silver), HG (Copper)
- Fixed income: ZB (30-yr T-Bond), ZN (10-yr T-Note), ZF (5-yr T-Note)
- FX futures: 6E (Euro), 6B (British Pound), 6J (Japanese Yen), 6A (Australian Dollar)
- Agricultural: ZC (Corn), ZS (Soybeans), ZW (Wheat)
- Crypto futures: BTC, ETH (per udcourse.com reference)

 *Volume Profile parameters (value_area_percent, HVN/LVN thresholds) may need instrument-specific tuning — ES has different volume distribution characteristics than CL or ZB.*

5.4 Revised Implementation Priority

Phase	Timeline	Key Additions from v2.0
Phase 0: Interfaces	Week 1	Add Volume Profile module headers; add orderflows.com indicator interface headers
Phase 1: Core Engine	Week 2–4	Tick ingestion, bar construction, delta engine, basic footprint — unchanged
Phase 2: Volume Profile Engine	Week 4–6	MOVED UP: Volume Profile is now Phase 2 (was Phase 3) — required as foundation for signal context
Phase 3: Order Flow Analytics	Week 6–8	Full imbalance suite, absorption, COT, trapped traders — now uses VP context
Phase 4: orderflows.com Signals	Week 8–10	Pulse, Turns, Ratio, Single Prints, Delta Surge implementations
Phase 5: VWAP + Advanced Signals	Week 10–12	VWAP engine; all VWAP strategies; Trader 7 signals subset
Phase 6: API Layer	Week 12–14	REST + WebSocket + VP API surface + orderflows.com signal endpoints
Phase 7: NinjaTrader + Testing	Week 14–18	NinjaTrader adapter; full test suite; historical replay
Phase 8: Security + Launch	Week 18–22	License engine; obfuscation; Paddle integration; NinjaTrader v1.0 public release

6. Additional Glossary Entries (v2.0)

Term	Definition	Source
Orderflows Pulse	Composite order flow indicator analyzing 7 variables (Order Flow, Delta, POC, Imbalances, Volume, Price Action, Swing Analysis) with multi-timeframe capability. Generates momentum jump, delta imbalance, and price reaction signals.	orderflows.com/pulse.html
Orderflows Turns	Order flow turning point detector analyzing 6 variables (POC, Bid/Ask volume, Volume, Price Action, Delta, Swing Analysis). Identifies defended/supported price areas for reversal entries.	orderflows.com/turns.html
Orderflows Ratio	Indicator detecting institutional-scale buyers (Bottom Heavy) at lows or sellers (Top Heavy) at highs — based on bid/ask ratio at price extremes.	orderflows.com/big-buyer-seller
Single Print / Last Buyer / Last Seller	Zero or near-zero volume at a bar extreme — signals the last trade at that price. Marks potential end of move. Orderflows implementation of Unfinished Business concept.	orderflows.com/last-buyerseller
Delta Surge	Free indicator detecting surges in delta trend. Works on both footprint and regular bar charts. Signals momentum acceleration.	orderflows.com/deltasurge
Bottom Heavy	Bar pattern where the majority of volume and bid/ask activity concentrates at the LOWER price levels — indicates strong passive or aggressive buying at lows.	orderflows.com/big-buyer-seller
Top Heavy	Bar pattern where the majority of volume and bid/ask activity concentrates at the UPPER price levels — indicates strong passive or aggressive selling at highs.	orderflows.com/big-buyer-seller
Momentum Jump	Pulse signal: sudden acceleration in order flow momentum at a specific price — identified as optimal entry point.	orderflows.com/pulse.html
Standalone Volume Profile	Volume Profile used as the primary trading instrument rather than a supporting indicator. Full analytical framework including profile types, shape classification, and 6 trading setups.	This spec §3; Dale VP book
D-Profile	Bell-curve shaped volume profile — balanced, rotational market with two-sided trade facilitation.	Dale Volume Profile book
P-Profile	Volume profile with concentration at upper prices (narrow base, wide top) — bullish accumulation/distribution pattern.	Dale Volume Profile book
b-Profile	Volume profile with concentration at lower prices (wide base, narrow top) — bearish distribution/accumulation pattern.	Dale Volume Profile book

Triple Confluence	Highest-conviction setup: Volume Profile key level + Order Flow signal (imbalance/absorption) + VWAP level all aligning at the same price.	Dale VWAP book; this spec §3.5
Value Area Re-Entry	Trade setup: price breaks out of the value area on weak/responsive delta and rapidly returns inside — fade the breakout.	This spec §3.5 Setup VP-4
Previous Session POC	The POC from the prior session that carries forward as a reference level for the current session — institutional memory level.	This spec §3.5 Setup VP-5

END OF SPECIFICATION — v2.0

40 total references | 4 PDFs | 12 orderflows.com pages | 1 YouTube channel | 23 web sources